

Higher Category Theory

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1 Background category theory

Definition 1.1 Given $F : \mathcal{C} \rightarrow \mathcal{D}$ and $x : \mathcal{D}$ we can define the slice category over F as the pullback:

$$\begin{array}{ccc} \mathcal{C}_{/F} & \longrightarrow & \mathcal{D}_{/x} \\ \downarrow & & \downarrow \\ \mathcal{C} & \longrightarrow & \mathcal{D} \end{array}$$

Explicitly objects are given as pairs $c : \mathcal{C}$ and morphisms $Fc \rightarrow x$.

Example 1.2 Over the Yoneda embedding $\mathfrak{y} : \mathcal{C} \rightarrow \hat{\mathcal{C}}$ and $F : \hat{\mathcal{C}}$ we have an element of $\mathcal{C}_{/F}$ is a pair of $c : \mathcal{C}$ and a morphism $\mathfrak{y}_c \rightarrow F$, that is an element $F(c)$.

Lemma 1.3 Given $F : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ the natural map $\text{colim}_{c \in \mathcal{C}_{/F}} \mathfrak{y}_c \rightarrow F$ is an equivalence

Proof. The natural map is given by choosing, for each $c : \mathcal{C}_{/F}$, the specified map $\mathfrak{y}_c \rightarrow F$.
We have equivalences, for each presheaf G

$$\text{hom}_{\hat{\mathcal{C}}}(\text{colim}_{c \in \mathcal{C}_{/F}} \mathfrak{y}_c, G) \cong \lim_{c \in \mathcal{C}_{/F}} \text{hom}(\mathfrak{y}_c, G) \cong \lim_{c \in \mathcal{C}_{/F}} G(c) \simeq \text{hom}(F, G)$$

giving the natural equivalence by Yoneda. The last equivalence is quick to check by hand. $\square$

Definition 1.4 (Kan extension (over inclusions)) Given $i : \mathcal{C}_0 \subseteq \mathcal{C}$, the inclusion of a subcategory, and $\mathcal{D}$ a (co)complete category, the restriction functor $i^* : \text{hom}(\mathcal{C}, \mathcal{D}) \rightarrow \text{hom}(\mathcal{C}_0, \mathcal{D})$ has left and right adjoints, given by

$$i_!(F)(x) = \text{colim}_{c \in \mathcal{C}_0/x} F(c)$$

$$i_*(F)(x) = \lim_{c \in \mathcal{C}_0} F(c)$$

1.1 Simplicial sets

We define the category Δ with objects given by $[n] = \{0, \dots, n\}$ and morphisms given by monotone maps. We define the category sSet to be presheaves on Δ . We denote the yoneda embedding of n to be Δ^n . This gives “expected” geometric results, such as $\text{hom}_{\text{sSet}}(\Delta^n, X) = X_n$, the n -simplices of X are given by morphisms of n -simplices.

Lemma 1.5 Given a simplicial set X , we have

$$X \cong \operatorname{colim}_{[n] \in \Delta/X} \Delta^n$$

Proof. This is exactly the fact that presheaves are colimits of representables. $\square$

Definition 1.6 For a simplicial set X we define

$$\pi_0^\Delta(X) = X_0 / \sim$$

where $x \sim y$ iff there exists $f : X_1$ such that $d_0(f) = x$ and $d_1(f) = y$.

Exercise 1 Find examples of simplicial sets where $\sim$ is not symmetric, or not transitive.

Solution For the first example we consider Δ^1 . This has non-degenerate 0-simplices given by $\Delta^1([0]) = [0] \rightarrow [1]$ by the constant maps at 0 and 1. And it has non-degenerate 1-simplex given by the identity map $(0, 1) : [1] \rightarrow [1]$. This has $d_0(0, 1) = 1$ and $d_1(0, 1) = 0$ so that $1 \sim 0$. But there is no 1-simplex giving $0 \sim 1$ so the relation isn't symmetric.

For a non-transitive example consider the spine $I^2 \subseteq \Delta^2$. This has non-degenerate 1-simplices given by $(0, 1)$ and $(1, 2)$. Hence we have $0 \sim 1$ and $1 \sim 2$ but not $0 \sim 2$.

Exercise 2 Let $X : \mathbf{sSet}$. Every simplex $x : X_n$ is uniquely determined by a surjection $\alpha : [n] \rightarrow [m]$ and $y : X_m$ with y non-degenerate and $\alpha^*(y) = x$.

Solution Note if x is degenerate, then by definition there exists some surjection α and some y such that $\alpha^*(y) = x$. Similarly if x is non-degenerate then there also exists a surjection, namely $\operatorname{id} : [n] \rightarrow [n]$, and this satisfies $\operatorname{id}^*(x) = x$.

We take the minimum $[m]$ such that there exists an α and a y . We claim such a y is non-degenerate. If it were then we would have a surjection $[k] \rightarrow [m]$ with $k < m$, composing would give a smaller value. Hence y is non-degenerate.

We now claim this is unique. Suppose we had $\beta : [n] \rightarrow [k]$ surjective with $\beta^*(z) = x$. Then by assumption $m \leq k$. By surjectivity we can construct a map $[k] \rightarrow [m]$ which composing with β gives α . Since z is not degenerate we deduce $k = m$ and β is the identity.

Exercise 3 Compute the limit and colimit of a simplicial set $X : \Delta^{\operatorname{op}} \rightarrow \mathbf{Set}$.

Solution Since the category $\Delta^{\operatorname{op}}$ has an initial object, we can compute that the limit of X is precisely X_0 . The colimit of X is given by the quotient of $\bigsqcup X_m$ by the relation generated by $x \sim \sigma^*(x)$. Note that all elements are related to an element of X_0 by the first vertex map. And pairs of elements of X_0 are related exactly when there is an $f : X_1$ connecting them. So we have that the colimit of X is precisely $\pi_0^\Delta(X)$.

Definition 1.7 In the category of topological spaces, we define $\Delta_{\operatorname{Top}}^n$ to be the subset of $\mathbb{R}^{n+1}$ of tuples who sum to 1. This defines a cosimplicial object in $\mathbf{Top}$ by

$$\begin{aligned} [n] &\mapsto \Delta_{\operatorname{Top}}^n \\ f : [n] &\rightarrow [m] \mapsto f_* \end{aligned}$$

where f_* is the unique affine linear extension of the map sending vertices of Δ_{Top}^n to their image under f . More precisely $f_*(t_0, \dots, t_n) = (v_0, \dots, v_m)$ where $v_m = \sum_{j \in f^{-1}(i)} t_j$

Definition 1.8 Given a topological space X , its singular simplicial set is given by

$$S(X) = [n] \mapsto \text{hom}(\Delta_{\text{Top}}^n, X)$$

We have a functor in the opposite direction given by Kan extension of the map sending $\Delta^n \mapsto \Delta_{\text{Top}}^n$. Explicitly

$$|X| = \text{colim}_{\Delta^n \in \Delta/X} \Delta_{\text{Top}}^n$$

We have $|_|$ is left adjoint to S .

Definition 1.9 1. $\partial\Delta^n$ has m -cells given by non-surjective $[m] \rightarrow [n]$.

2. Given $S \subseteq [n]$ the **horn** Λ_S^n has m -cells given $f : [m] \rightarrow [n]$ so there is $i \in [n] \setminus S$ s.t. $i \notin \text{im}(f)$.

3. The **spine** I^n has m -cells given by $[m] \rightarrow [n]$ with image $\{j\}$ or $\{j, j+1\}$.

Definition 1.10 We can form the subcategory $\Delta_{\leq n} \subseteq \Delta$. The natural adjunction given by kan extending the inclusion gives monads on simplicial sets called the skeleta and coskeleta. Specifically

$$\text{sk}_n(X) = i_! i^* X$$

$$\text{cosk}_n(X) = i_* i^* X$$

We have $\text{cosk}_n(X)_k = \text{hom}(\text{sk}_n(\Delta^k), X)$

Exercise 4 For all $n \geq 0$ we have a pushout

$$\begin{array}{ccc} \bigsqcup_{J_n} \partial\Delta^n & \longrightarrow & \text{sk}_{n-1}(X) \\ \downarrow & & \downarrow \\ \bigsqcup_{J_n} \Delta^n & \longrightarrow & \text{sk}_n(X) \end{array}$$

Where J_n are the non-degenerate n -simplices.

Solution We will turn this into a Yoneda style gluing argument to make our lives easier. Let Y be a simplicial set. We have to show that

$$\begin{array}{ccc} \text{hom}(\bigsqcup_{J_n} \partial\Delta^n, Y) & \longleftarrow & \text{hom}(\text{sk}_{n-1}(X), Y) \\ \uparrow & & \uparrow \\ \text{hom}(\bigsqcup_{J_n} \Delta^n, Y) & \longleftarrow & \text{hom}(\text{sk}_n(X), Y) \end{array}$$

is a pullback. By using the adjunction, and the Yoneda lemma this diagram is equivalent to

$$\begin{array}{ccc} \prod_{J_n} \text{hom}(\partial\Delta^n, Y) & \longleftarrow & \text{hom}(i_{n-1}X, i_{n-1}Y) \\ \uparrow & & \uparrow \\ (Y_n)^{J_n} & \longleftarrow & \text{hom}(i_nX, i_nY) \end{array}$$

Further we note that a map $\partial\Delta^n \rightarrow Y$ is determined by its values on the $n-1$ cells of $\partial\Delta^n$, realising this set as a subset of $\prod_{i=0}^n Y_{n-1}$.

Now we can explicitly describe the maps in this diagram. $\text{hom}(i_nX, i_nY)$ are given by natural transformations, collections of $f_i : X_i \rightarrow Y_i$ for $0 \leq i \leq n$.

- The map $\text{hom}(i_nX, i_nY) \rightarrow \text{hom}(i_{n-1}X, i_{n-1}Y)$ sends $(f_0, \dots, f_n)$ to $(f_0, \dots, f_{n-1})$.
- The map $\text{hom}(i_nX, i_nY) \rightarrow (Y_n)^{J_n}$ sends a tuple f to the map $\lambda\sigma.f_n(\sigma)$.
- The map from $(Y_n)^{J_n}$ sends g to the map $\lambda(\sigma, i).d_i g(\sigma)$.
- The top map sends a tuple f to the map $\lambda(\sigma, i).f_{n-1}(d_i\sigma)$.

Thus to check this diagram is a pullback we need to show the following. Suppose we have maps $f : \text{hom}(i_{n-1}X, i_{n-1}Y)$ and $g : (Y_n)^{J_n}$ which satisfy for all $\sigma : J_n$ and i that $d_i g(\sigma) = f_{n-1}(d_i\sigma)$. Then we must construct a unique map $\text{hom}(i_nX, i_nY)$ mapping to f and g . We do this by constructing f_n . If $x : X_n$ then either $x \in J_n$ in which case $f_n(x) := g(x)$, Or x is degenerate and $x = \alpha^*(y)$ for some $y : X_m$ with $m < n$. Then define $f_n(x) = \alpha^*(f_m(y))$. This is well defined, natural and unique.

Definition 1.11 Given a category $\mathcal{C}$, its nerve is the simplicial set given by

$$[n] \mapsto \text{hom}([n], \mathcal{C})$$

That is chains of n composable maps.

Definition 1.12 Given a group G its classifying space is the geometric realisation of its nerve as a category with one object.

Definition 1.13 A **Kan complex** is a simplicial set X such that for all Horns Δ_j^n there are solutions to lifting problems

$$\begin{array}{ccc} \Lambda_j^n & \longrightarrow & X \\ \downarrow & \nearrow & \\ \Delta^n & & \end{array}$$

Exercise 5 Given a simplicial set X show that for all $n \geq 0$ that $\text{cosk}_n(X)$ is a Kan complex. Further show the natural map $X \rightarrow \text{cosk}_n(X)$ induces isomorphisms on π_k for $k < n$ and that $\pi_k(\text{cosk}_n(X)) = 0$ for $k \geq n$.

Solution Suppose we have a map $\Lambda_j^m \rightarrow \text{cosk}_n(X)$. By adjunction this is equivalently given by a map $\text{sk}_n(\Lambda_j^m) \rightarrow X$. Note that for $m > n + 1$ the map $\text{sk}_n(\Lambda_j^m) \rightarrow \text{sk}_n(\Delta^m)$ is just the Horn inclusion. Hence we have a lift. If $m \leq n$ then the inclusion becomes an isomorphism, so therefore has a lift. In the case $m = n + 1$ we have $\text{sk}_n(\Lambda_j^{n+1}) \rightarrow \text{sk}_n(\Delta^{n+1})$ becomes the inclusion $\Lambda_j^{n+1} \rightarrow \partial\Delta^{n+1}$. We can fill against this by filling two a the whole simplex, and restricting.

This same argument makes the isomorphism of homotopy groups clear.

Example 1.14 The singular simplicial set of a topological space is a Kan complex. This follows quickly by adjunction, noting that a horn is a retract of Δ^n in topological spaces.

Example 1.15 The nerve of a category is a Kan complex if and only if the category is a groupoid.

Definition 1.16 Given $f, g : X \rightarrow Y$ maps of simplicial sets, we say they are homotopic if there exist a map $H : \Delta^1 \times X \rightarrow Y$ restricting to f, g on boundary. We can define the simplicial homotopy groups as homotopy classes of maps $(\Delta^n, \partial\Delta^n) \rightarrow (X, x)$.

2 ∞ -Categories

We will start with a definition which is not of an ∞ -category, but a weaker notion. In particular these will be a simplicial set with composition, but the space of compositors isn't necessarily contractible.

Definition 2.1 A **composer** is a simplicial set with the extension property against spine inclusions $I^n \rightarrow \Delta^n$.

Example 2.2 The singular set of a topological space is a composer. Given paths $p_i : x_i \rightarrow x_{i+1}$, the data of a spine map, we define their composition as the composition of the paths. If you aren't convinced this gives an actual n -cell, there is some way to see this either purely topologically, or by inductively filling 2-horns, 3-horns, etc. until you get a full filler of Δ^n , using

Definition 2.3 Two 1-simplices $f, g : x \rightarrow y$ are **equivalent** if there is a two simplex $\sigma : \Delta^2 \rightarrow X$ such that

$$\sigma_{0,1} = f$$

$$\sigma_{0,2} = g$$

$$\sigma_{1,2} = \text{id}_y$$

Definition 2.4 Let X be a simplicial set. Define the **homotopy category** hX to be the category with:

1. Objects: X_0

2. Morphisms: freely generated by X_1 . Write the free composition by $f * g$.

quotiented by the relations

1. $s_0(x) = \text{id}_x$
2. For every 2-simplex Σ with boundary (f, g, h) we have $h = g * f$.
3. If $f \sim f'$ then $f * g \sim f' * g$ and $g' * f \sim g' * f'$.

This construction gives a functor $h : \mathbf{sSet} \rightarrow \mathbf{Cat}$.

In a composer, the free generation collapses - every formal composite will be equivalent to a single morphism.

Lemma 2.5 *Suppose X is a composer which lifts wrt inner 3-horn inclusions. Let f and g be composable morphisms in X . Then*

1. *There exists a composite of f and g*
2. *The relation of equivalence is an equivalence relation*
3. *Any two composites of f and g are equivalent*
4. *given a 2-simplex σ with $\sigma = (\text{id}_x, h, h')$ then $h \sim h'$*

Proof. This is an exercise in horn filling! □

Furthermore in such a composer we have a nicer description of its homotopy category: It is given by the category $\pi(X)$ with objects X_0 and morphisms equivalence classes of X_1 . Associativity of composition also follows from horn filling.

Definition 2.6 A simplicial set X is an ∞ -category if it has fillers for all inner horn inclusions. A functor between ∞ -categories is a map of simplicial sets.

Definition 2.7 A morphism in an ∞ -category is an **equivalence** if its image in the homotopy category is an isomorphism. Equivalently there are two cells σ and τ witnessing whose outer cells witness that f has a right and left inverse.

Definition 2.8 Given an ∞ -category we construct its maximal subgroupoid as a pullback of simplicial sets

$$\begin{array}{ccc} \mathcal{C}^\simeq & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ N(h\mathcal{C}^\simeq) & \longrightarrow & N(h\mathcal{C}) \end{array}$$

2.1 Examples of infinity categories

Apart from the usual examples, to cook up more examples we will use simplicially enriched categories.

Definition 2.9 A **simplicial category** is a category enriched in simplicial sets, with the cartesian monoidal structure.

Explicitly, we have a set $\mathcal{C}$ of objects, and for each $x, y : \mathcal{C}$ a simplicial set of morphisms $\mathcal{C}(x, y)$, with identities and composition.

By abstract nonsense the category of small simplicial categories is bicomplete. Additionally noting that the maps $c : \text{Set} \rightarrow \text{sSet}$ and $\pi_0, \text{ev}_0 : \text{sSet} \rightarrow \text{Set}$ are monoidal, we obtain maps

$$c : \text{Cat} \rightarrow \text{Cat}_\Delta$$

$$\pi, u : \text{Cat}_\Delta \rightarrow \text{Cat}$$

These form an adjoint triple $\pi \dashv c \dashv u$

Definition 2.10 A simplicial functor $f : \mathcal{C} \rightarrow \mathcal{D}$ of simplicial categories is a **weak equivalence** if it is a weak equivalence on hom simplicial sets, and $\pi(f) : \pi(\mathcal{C}) \rightarrow \pi(\mathcal{D})$ is essentially surjective. This is the “weak” version of being fully faithful and essentially surjective.

Definition 2.11 A morphism f in a simplicial category $\mathcal{C}$ is a zero cell of $\mathcal{C}(x, y)$. It is an **equivalence** if $\pi(f)$ is an isomorphism.

Now we would like to perform the nerve construction on simplicial categories. For this we need to define the simplicially enriched version of $[n]$.

Definition 2.12 Let J be a finite non-empty linearly ordered set and $i, j : J$. Let $P_{i,j} := \{I \subseteq J \mid i, j \in I \text{ and } k \in I \Rightarrow i \leq k \leq j\}$. This is partially ordered by inclusion. If $i \leq j \leq k$ there is a canonical map

$$P_{i,j} \times P_{j,k} \rightarrow P_{i,k}$$

given by the union.

We can use this to define a simplicial category $\mathcal{C}[\Delta^J]$ with

1. Objects J
2. morphisms $i \rightarrow j$ given by the simplicial set $N(P_{i,j})$.

Composition is induced by the canonical map.

We can explicitly describe these homsets, since for all n we have $P_{0,n} \simeq [1]^{n-1}$, by taking the indicator of the subset. We can also see $P_{i,j} \simeq P_{0,j-i}$.